

# Enhanced co-pyrolysis of corn stalk and bio-tar into phenolic-rich biooil: Kinetic analysis and product distributions

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## ARTICLE INFO

### Keywords:

Corn stalk

Bio-tar

Pyrolysis kinetics

Synergetic effect

Phenolic compounds

## ABSTRACT

This study explores the interaction between corn stalk and bio-tar through co-pyrolysis, aiming to provide insights for the valorization of bio-tar and biomass into phenolic-rich biooil. The study focuses on the thermal behavior analysis, pyrolysis kinetic, and product distributions during the co-pyrolysis process. The results demonstrate a significant interaction between bio-tar and corn stalk during the co-pyrolysis process. Notably, the co-pyrolysis of bio-tar and corn stalk with a ratio of 1:3 exhibits the most pronounced synergistic effect, enhancing the thermal decomposition of corn stalks. Kinetic modeling calculations reveal a reduction in activation energy for all mixing ratios, with a substantial decrease of 72.87 kJ/mol observed for the 1:3 bio-tar-to-stalk ratio. Additionally, the distribution of pyrolysis products varies with different mixing ratios, with lower bio-tar addition (ratio of bio-tar to stalk = 1:3) favoring phenolic compound production (>70% of phenols in biooil). These findings contribute to a better understanding of the interaction between bio-tar and corn stalk during co-pyrolysis, offering valuable insights for the efficient utilization of biomass resources.

## 1. Introduction

In the context of carbon neutrality goals in 2060, exploring low-carbon and sustainable energy and material alternatives has become an urgent task. The conversion and utilization of biomass resources into liquid fuels and functional materials have gained considerable attention due to their advantages, such as renewability and carbon neutrality [1–3]. Particularly, biochar has a wide range of applications in soil improvement, water purification, agriculture, and environmental protection, making it a current research hotspot, owing to its porous nature, high stability, strong adsorption characteristics, and carbon sequestration benefits [4–6]. However, the production of biochar through biomass pyrolysis and carbonization unavoidably yields a byproduct known as bio-tar [7]. Bio-tar is a complex mixture primarily composed of numerous heavy aromatic compounds and their derivatives, which exhibits high viscosity and can cause pipeline blockages while posing potential hazards to the environment [8–10]. Consequently, the effective treatment of bio-tar is a pressing issue that demands immediate

attention.

Nowadays, bio-tar treatment technologies include direct combustion, catalytic pyrolysis, thermal plasma, hydrogenation reforming, and distillation technologies. Direct combustion involves oxidizing bio-tar in the presence of air, effectively utilizing the heating value of bio-tar but producing polluting gases such as CO and NO<sub>x</sub>, which can cause environmental pollution [11]. Catalytic pyrolysis refers to the conversion and removal of volatile components in bio-tar through catalytic action under thermal atmosphere. For example, Li et al. [12] used Ni/HZSM-5 catalyst to pyrolyze wheat straw tar, achieving a tar removal rate of up to 89.34% at 800 °C. Zhang et al. [13] discovered that pine wood activated biochar achieved 92.6% bio-tar removal and its phenolic compounds were clear out. Thermal plasma removal offers the advantages of simplicity in structure and complete degradation, rapidly converting bio-tar into gases such as H<sub>2</sub>, CO, and CO<sub>2</sub> [14–16]. Hydrogenation reforming can achieve the upgrading of bio-tar and be used to prepare high-quality biofuels rich in stable compounds (such as aromatics, hydrocarbons and alkyl phenols), realizing the resource utilization of

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bio-tar. To be specific, Lee et al. [17] studied the in-situ catalytic hydrogenation and deoxygenation of bio-tar and successfully prepared high quality biofuels. Distillation technology can clear out and separate bio-tar, while phenol oil, naphthalene oil, light wash oil fraction, and heavy oil can be separated according to different distillate temperatures. Vivanpatarakij et al. [18] achieved complete removal of bio-tar using a bio-tar steam reformer. Among them, co-pyrolysis technology can regulate product distribution through the interaction of raw materials [19], especially the co-pyrolysis of lignocellulose and carbon materials shows significant synergistic effects and could improve the quality of liquid [20,21]. Corn stalk contains components such as cellulose, hemicellulose, and lignin, which can be decomposed compounds such as acids, furans, ketones, and phenols [20–22]. These compounds may react with phenols, aldehydes, esters, and hydrocarbons in bio-tar, possible to improve the distribution and quality of the products [23]. It is an insightful way for the resource utilization of bio-tar and corn straw via co-pyrolysis, not only helping alleviate environmental problems but also reducing the reliance on fossil fuels[1].

This study proposes a comprehensive utilization strategy for the co-pyrolysis of corn stalk and the bio-tar derived from its biochar production process. Firstly, the interaction between corn stalk and bio-tar is analyzed through thermogravimetric-differential thermogravimetric (TG-DTG) experiments. Secondly, the synergistic effect between corn stalk and bio-tar is further investigated from the perspective of activation energy using different pyrolysis kinetics models (including DAEM, Friedman and FWO). Furthermore, the influence of incorporating stalk on the pyrolysis products of bio-tar is examined by investigating the distribution and composition of the pyrolysis experimental products. The primary objective of this study is to comprehensively investigate the interaction between corn stalk and bio-tar under co-pyrolysis conditions and provide valuable insights for the efficient utilization of bio-tar.

## 2. Materials and methods

### 2.1. Materials

The corn stalk and pyrolysis bio-tar used in this study were obtained from Nanjing Qinfeng Stalk Technology Co., Ltd, Jiangsu Province, CHINA. Prior to the pyrolysis experiment, corn stalk was crushed and sieved to a particle size of 60–100 meshes. The pyrolysis bio-tar was dried in an oven at 105 °C for 48 h and then grinded to particles within the 0.1–0.3 mm ranges. The elemental compositions of bio-tar and corn stalk were detected by an inductively coupled plasma emission spectrometer (ICP, Optima 5300DV, Perkin Elmer, USA). ASTM E870–82 (2006) was used to conduct the proximate analysis of bio-tar and corn stalk[24]. Table 1.

### 2.2. Thermogravimetric analysis

TGA (Thermogravimetric Analysis) was performed by NETZSCH STA 449F3 equipment with N<sub>2</sub> (99.999%) as carrier gas at a flow rate of 20 mL/min. The heating rates were controlled at 10, 20, and 30 K/min. In

**Table 1**

The composition analysis of corn stalk and bio-tar.

|            | Ultimate analysis (wt%) <sup>a</sup>  |               |      |      |                                |
|------------|---------------------------------------|---------------|------|------|--------------------------------|
|            | C                                     | H             | N    | S    | O <sup>c</sup>                 |
| Corn stalk | 38.36                                 | 5.86          | 1.21 | 0.15 | 45.16                          |
| Bio-tar    | 70.30                                 | 6.27          | 2.17 | 0.34 | 13.09                          |
|            | Proximate analysis (wt%) <sup>b</sup> |               |      |      | Fixed carbon (FC) <sup>d</sup> |
|            | Ash (A)                               | Volatiles (V) |      |      |                                |
| Corn stalk | 9.26                                  | 64.05         |      |      | 26.68                          |
| Bio-tar    | 7.83                                  | 67.32         |      |      | 24.85                          |

a) Daf, dry air and ash free basis; b) Db, dry air basis; c) Calculated by difference, O= 100-C-H-N-S-A; d) Calculated by difference, FC= 100-A-V.

each experiment, the weight of sample was approximately 50 mg and loaded into an Al<sub>2</sub>O<sub>3</sub> crucible, and the temperature range selected for the analysis was from room temperature to 800 °C. TG represents mass loss, and DTG represents the rate of mass loss.

### 2.3. Kinetic analysis

The pyrolysis kinetic analysis were calculated based on the Distribution of activation energy model [25–30] (DAEM), Flynn-Wall-Ozawa [31–36] (FWO), Kissenger-Akahira-Sunose [37–42] (KAS). It is primarily based on the following assumptions: 1. The reaction system is composed of countless independent first-order reactions, each with a different activation energy. 2. The activation energies of the individual reactions exhibit a continuous distribution function [43].

The above models were based on the Arrhenius formula (1):

$$k = k_0 e^{-E/RT} \quad (1)$$

k is the rate constant, k<sub>0</sub> is the pre-exponential factor, E is the activation energy, R is the gas constant (8.314 J·mol<sup>-1</sup>·K<sup>-1</sup>) and T is the reaction temperature (K).

Furthermore, the calculation method of conversion rate (Eq. (2)) can be obtained after derivation of Eq. (1).

$$\alpha(T) = \int_0^{+\infty} (1 - \exp(-\frac{k_0}{\beta} \int_{T_0}^T \exp(-\frac{E}{RT}) dT) f(E) dE \quad (2)$$

where conversion is defined as Eq. (3)

$$\text{Converions rate}(\%) = \alpha(T) = \frac{M_T - M_0}{M_\infty - M_0} \quad (3)$$

where M<sub>T</sub> is the mass percentage corresponding to T(K), M<sub>0</sub> and M<sub>∞</sub> are the mass percentage corresponding to the initial and end reaction temperature, respectively. T<sub>0</sub> and T are the initial and reaction temperature. f(E) is activation energy distribution function. β is heating rate.

The derivative of T on both sides of Eq. (2) can get Eq. (4).

$$\frac{d\alpha}{dT} = \int_0^{+\infty} \frac{k_0}{\beta} \exp(-\frac{E}{RT}) - \frac{k_0}{\beta} \int_{T_0}^T \exp(-\frac{E}{RT}) dT f(E) dE \quad (4)$$

where  $\frac{d\alpha}{dT}$  is conversion rate. f(E) can be calculated by Eq. (5).

$$\int_0^\infty f(E) dE = 1 \quad (5)$$

1) For DAEM as shown in Eq. (6).

$$\ln\left(\frac{\beta}{T^2}\right) = \ln\left(\frac{k_0 R}{E}\right) + 0.607 - \frac{E}{RT} \quad (6)$$

2) For Friedman as show in Eq. (7).

$$\ln\left(\beta \frac{d\alpha}{dT}\right) = \ln(Af(\alpha)) - \frac{E}{RT} \quad (7)$$

3) For FWO analysis as shown in Eq. (8).

$$\ln\beta = \ln\left(\frac{AE}{Rf(\alpha)}\right) - 1.052 \frac{E}{RT} - 5.331 \quad (8)$$

To deeply study the distribution of activation energy, the theoretical activation energy can be calculated by Eq. (9).

$$E_{Cal} = mE_{\text{corn stalk}} + (1 - m)E_{\text{bio-tar}} \quad (9)$$

Where m is the percentage of corn stalk in the mixture, E<sub>Cal</sub> is the theoretical activation energy, E<sub>corn stalk</sub> and E<sub>bio-tar</sub> are the activation energies for pure stalk and bio-tar, respectively.

## 2.4. Bench-top pyrolysis experiments

For each pyrolysis experiment, 0.3 g sample was weighed using a microbalance and placed in a hanging basket. The weights of the quartz tube and condenser tube were recorded. The reaction system was assembled, and the condenser tube was immersed in liquid nitrogen for product collection. To minimize the impact of residual air, nitrogen gas was passed through the system at 200 mL/min for 5 min. The rapid pyrolysis temperature was selected as 600 °C based on thermogravimetric analysis and to ensure complete volatilization of the raw material. After the gas flow ceased, the hanging basket was quickly placed in the center of the reactor and weighed after cooling to room temperature. The reaction time was controlled at 20 min. The flow rate of nitrogen was 200 mL/min throughout the entire reaction process. The gas was recaptured via a gas balloon connected to the condenser tube. Liquid products were analyzed using gas chromatography-mass spectrometry (Agilent 7890-5977B, Santa Clara, CA) with an HP-5 MS capillary column (30 m × 0.25 mm × 0.25 mm). Fig. 1 displayed the detailed experimental procedure. The yield of biochar, biooil, and gases can be calculated according to Eqs. (10) - (12). Each experimental group was conducted in duplicate, with the error not exceeding 5%.

$$\text{Biochar yield}(\%) = \frac{\text{Mass}_{\text{biochar}}}{\text{Mass}_{\text{feedin}}} \times 100\% \quad (10)$$

$$\text{Biooil yield}(\%) = \frac{\text{Mass}_{\text{biooil}}}{\text{Mass}_{\text{feedin}}} \times 100\% \quad (11)$$

$$\text{Gases yield}(\%) = (1 - \text{biochar yield} - \text{biooil yield}) \times 100\% \quad (12)$$

$$Y_{\text{Cal}} = nY_{\text{pure corn stalk}} + (1 - n)Y_{\text{pure bio-tar}} \quad (13)$$

The value of cal. was determined by Eq. (13). Where  $n$  represents the corresponding proportion of stalk in the mixture, and  $Y_{\text{Cal}}$  is the calculated theoretical yield, and  $Y_{\text{exp}}$  is the yield obtained under experimental conditions.

## 3. Results and discussions

### 3.1. TG-DTG analysis

To analyze the interaction relationship between different mixing ratios of stalk and bio-tar, thermogravimetric analysis was conducted on pure bio-tar, pure corn stalk, and the mixtures with different mixing ratios, under a heating rate of 10 K/min (as shown in Fig. 2. 3). The thermal decomposition of pure corn stalk can be roughly divided into

four stages: the drying stage (room temperature to 110 °C, primarily involving the removal of water), the initiation stage (110 to 210 °C, with a slow decomposition rate and generation of small molecular gases such as H<sub>2</sub>, CO, and CO<sub>2</sub>), the main decomposition stage (210 to 370 °C, involving the decomposition of hemicellulose and cellulose), and the continuous decomposition stage (370 to 800 °C, involving the decomposition of lignin), which was consistent with the findings reported by Zou et al. [25]. In addition, the thermal decomposition of bio-tar can be roughly divided into the drying stage (room temperature to 150 °C), the pyrolysis stage (150 to 580 °C), and the carbonization stage (580 to 800 °C). Firstly, the weaker C-C or C-O bonds in the branched chains of polycyclic organic substances decomposed to form light bio-tar [44]. Subsequently, with the temperature increasing, the weaker bonds in the side chains of light components (such as methoxy groups) further was broken to form the final pyrolysis small molecule products (phenol, hydrocarbon, furan, ketones, etc.) [12].

Moreover, the thermogravimetric curves of the different mixing ratios exhibited similar stage characteristics, indicating a certain degree of interaction between the two components. The mixed samples showed a deeper extent of pyrolysis reaction, promoting the decomposition of more volatile components. In terms of reaction rates, the pure stalk exhibited the highest maximum reaction rate, while the bio-tar to stalk= 1:3 mixture exhibited the lowest maximum reaction rate among the mixed samples. This can be attributed to the broadened temperature range for reactions in the mixed samples, resulting in a decrease in the reaction rate. Overall, the interaction between the two feedstocks facilitated mass loss while reducing the rate of mass loss, with the bio-tar to stalk= 1:3 mixture showing the most prominent effect.

### 3.2. Thermogravimetric synergistic effect analysis

This study further investigated the thermogravimetric (TG) curves of bio-tar and corn stalk at different mixing ratios, aiming to explore the interaction between the two components through calculating theoretical values and comparing them with experimental values. The research findings demonstrated that the experimental groups under all three mixing conditions showed a significant increase in mass loss compared to the single-component groups, indicating that the interaction between the two components further promotes the decomposition of volatile matter and increases mass loss. Under different mixing ratios, there were varying degrees of advancement or delay in the temperature range of the main reaction. Specifically, the experimental group of bio-tar to stalk ratio of 3:1 exhibited a slightly delay distribution of the main peak, which may be attributed to the aggregation of bio-tar into solid chunks at high temperatures, impeding the decomposition of corn stalk. The experimental group of bio-tar to stalk ratio of 1:1 generally aligned with the theoretical group in terms of the distribution of the main peak. However, the experimental group of bio-tar to stalk ratio of 1:3 showed a significant forward shift of the main peak, indicating that the reaction at a lower temperature promotes the occurrence of the pyrolysis process for this mixing ratio, where a small amount of bio-tar blending enhanced the distribution of the pyrolysis behavior. When comparing the synergistic effect of the three mixing ratios, it was found that the bio-tar to stalk ratio of 1:3 exhibited the greatest difference in weight loss, indicating the most significant synergistic effect. Regarding the corresponding temperatures of the main peaks, the bio-tar to stalk ratio of 3:1 exhibited a slight backward shift, while the bio-tar to stalk ratio of 1:3 showed a significant decrease, indicating that a lower proportion of bio-tar blending could promote the pyrolysis process. In terms of the maximum weight loss rate, all three mixing ratios showed varying degrees of reduction, with the bio-tar to stalk ratio of 1:3 experiencing the most significant decrease. However, with the forward shift of the main peak distribution, a portion of the reactive components participated in the reaction at an earlier stage, leading to a decrease in the weight loss rate. In summary, the bio-tar to stalk ratio of 3:1, 1:1, and 1:3 all had a certain promoting effect on mass loss, with the bio-tar to stalk ratio of 1:3

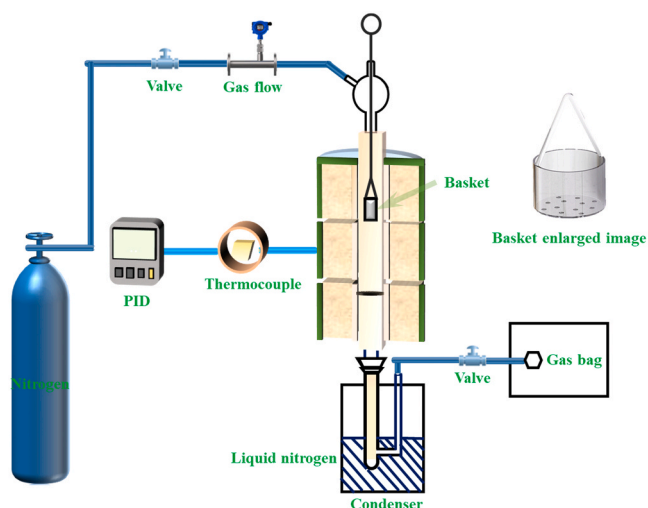


Fig. 1. The experimental procedure for bench-top pyrolysis experiment.

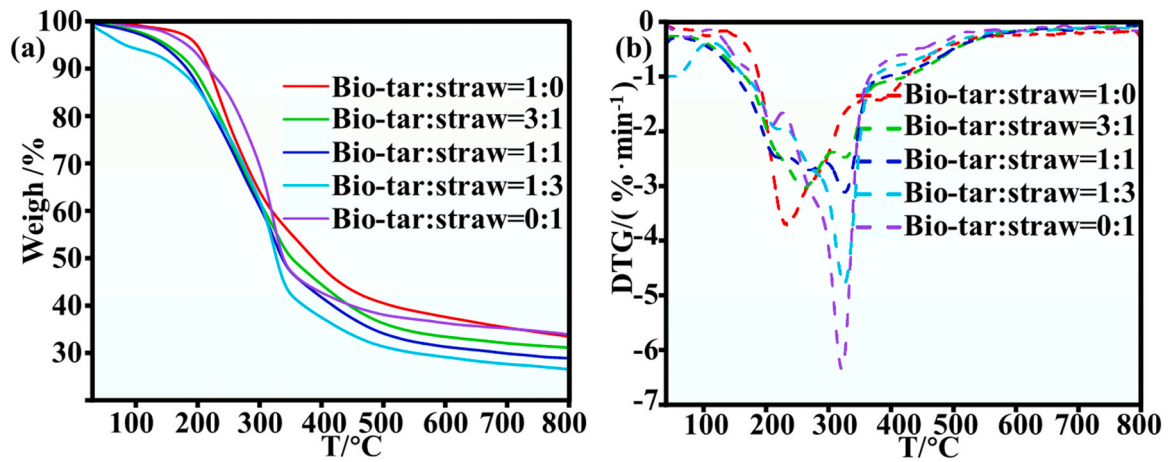


Fig. 2. The TG-DTG curves of bio-tar and corn stalk.

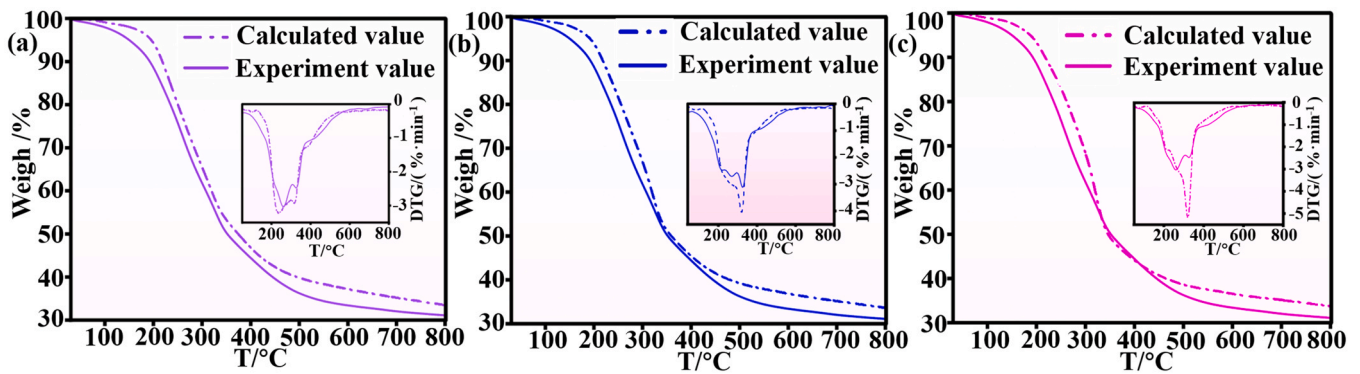


Fig. 3. The synergistic effect between different mixing ratios of bio-tar to corn stalk (a. 3:1; b. 1:1; and c. 1:3).

displaying the most significant synergistic effect and forward shift of the main peak. This highlighted the characteristics and advantages of the copyrolysis process and demonstrated the potential for utilizing the interaction between these two components to better facilitate the conversion of raw materials. Therefore, to further reveal the reasons behind the synergistic effect and forward shift of the main peak, this study would focus on the bio-tar-to stalk ratio of 1:3 as a representative case and conduct further exploration from a kinetic perspective. Table 2.

### 3.3. Pyrolysis kinetic analysis

Fig. 4 illustrates the thermal gravimetric curves of corn stalk (a), bio-tar (b), and the 1:3 mixture (c) under different heating rates, along with the disparity curve between theoretical and experimental data (d). The results indicate that higher heating rates led to a greater mass loss rate and delayed peak distribution temperature for all three samples, primarily influenced by heat and mass transfer effects and their hysteresis. The heating rate had a negligible impact on the final difference in mass loss or volatile yield. The thermal gravimetric disparity demonstrated a significant promotion effect of different heating rates on the 1:3 mixture,

particularly at 10 K/min where the disparity reaches its maximum value of 7.91%. This highlighted the relevance of heating rate (heat and mass transfer efficiency) in the interaction between bio-tar and corn stalk components. The slower pyrolysis rate enhanced uniform heat transfer, facilitating the complete reaction.

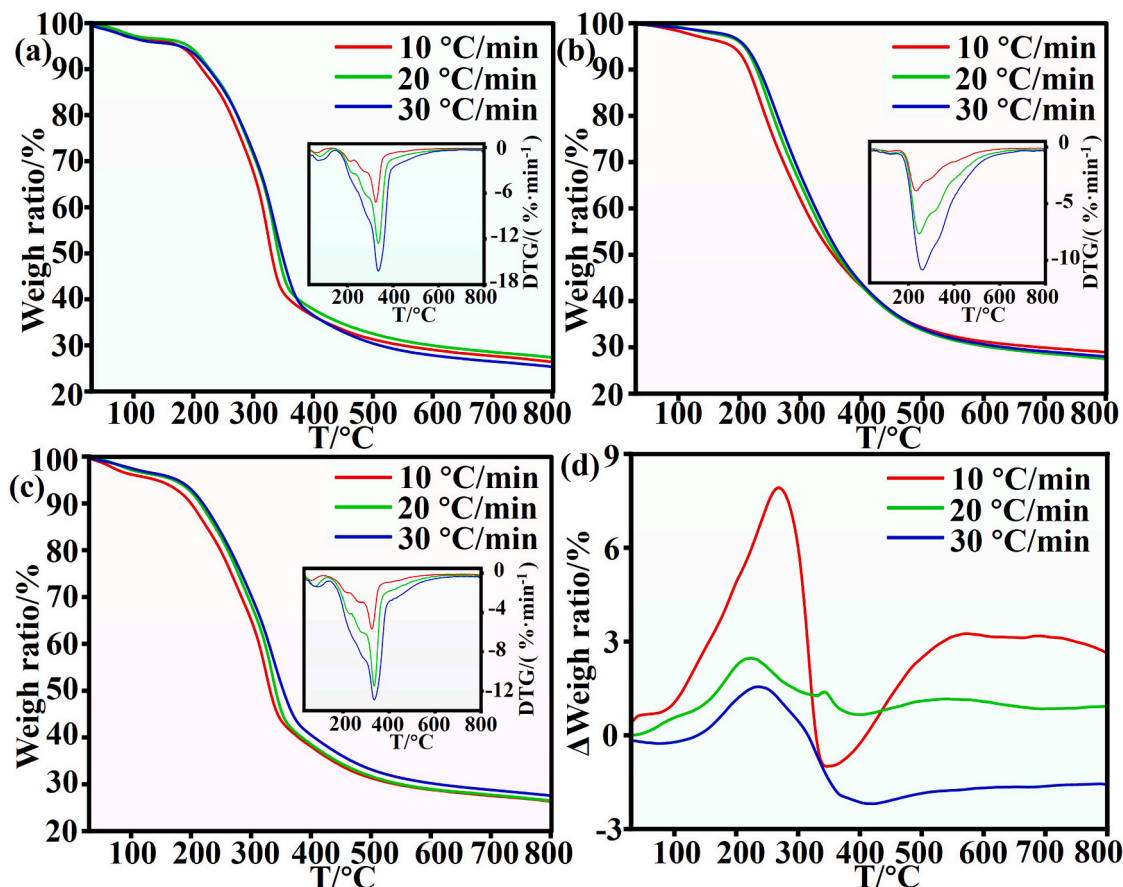
Furthermore, the synergistic effects of a 1:3 ratio of bio-tar to corn stalk were investigated using three models, namely DAEM, Friedman, and FWO. It was found that the results from DAEM and FWO were in agreement, while Friedman showed some discrepancies, possibly due to its utilization of differential transformation rates, which rendered it sensitive to numerical differences. The  $R^2$  values for all three models exceeded 0.900, with a maximum of 1.000, indicating their good adaptability and accurate reflection of changes in activation energy. It was observed that the activation energy of corn stalk was significantly higher than that of bio-tar, while the activation energy for the 1:3 mixture fell within the range between the two constituents. The gradual increase followed by a decrease in the activation energy of corn stalk, exemplified by the DAEM model, was observed as it rose from 173.42 kJ/mol to 211.16 kJ/mol and subsequently decreased to 169.09 kJ/mol. On the other hand, the activation energy of bio-tar

**Table 2**  
Comparison of difference parameters of samples with different mixing ratios.

| Sample                   | Max. $\Delta^a$ (%) | Theor. max. loss tem. (°C) | Actual Max. loss tem (°C) | Theor. max. rate (°C·min <sup>-1</sup> ) | Actual max. rate (°C·min <sup>-1</sup> ) |
|--------------------------|---------------------|----------------------------|---------------------------|------------------------------------------|------------------------------------------|
| Bio-tar: Corn stalk= 3:1 | 2.44                | 238.27                     | 260.16                    | 3.23                                     | 3.02                                     |
| Bio-tar: Corn stalk= 1:1 | 2.55                | 317.74                     | 325.47                    | 4.04                                     | 3.10                                     |
| Bio-tar: Corn stalk= 1:3 | 2.66                | 319.32                     | 260.16                    | 5.19                                     | 3.03                                     |

a:  $\Delta$  = actual mass loss -theoretical mass loss.





**Fig. 4.** TG curves of pure corn stalk (a), pure bio-tar (b) and mixture of bio-tar and corn stalk (1:3) and its difference between theoretical and experimental curves (d) at different heating rates (10 K/min, 20 K/min and 30 K/min).

exhibited a relatively uniform distribution, except for higher values observed at conversion rates of 70% and 80%, with the majority falling within the range of 113.00 to 159.00 kJ/mol. These observations can be attributed to the complex structure of cellulose, hemicellulose, and lignin present in corn stalk, as well as the simpler and more dispersed aromatic compounds, such as phenols, found in bio-tar. [Table 3](#).

To further investigate the adaptability of the three models (DAEM,

Friedman, and FWO) to the 1:3 ratio of bio-tar to corn stalk, the fitting performance of these models was compared in [Fig. 5](#). The results demonstrated that both DAEM and FWO exhibited a positive correlation between their slopes and activation energy, with a relatively concentrated distribution of activation energy. However, these methods corresponded to lower temperature distributions at lower conversion rates, exhibiting a trend from the upper right to the lower left in the graph as the

**Table 3**  
Comparison of activation energy and correlation coefficient ( $R^2$ ) under three models.

|                           | Convers. Rate (%) | DAEM<br>(kJ/mol) | $R^2$ | Friedman<br>(kJ/mol) | $R^2$ | FWO<br>(kJ/mol) | $R^2$ |
|---------------------------|-------------------|------------------|-------|----------------------|-------|-----------------|-------|
| Corn stalk                | 20                | 173.42           | 0.901 | 151.41               | 0.842 | 173.09          | 0.910 |
|                           | 30                | 202.71           | 0.990 | 204.67               | 0.929 | 201.38          | 0.991 |
|                           | 40                | 210.97           | 0.998 | 239.34               | 1.000 | 209.59          | 0.998 |
|                           | 50                | 211.16           | 0.987 | 250.88               | 0.969 | 210.05          | 0.988 |
|                           | 60                | 193.09           | 0.999 | 236.75               | 0.975 | 193.08          | 0.999 |
|                           | 70                | 229.63           | 0.988 | 225.69               | 0.972 | 164.16          | 0.990 |
|                           | 80                | 169.09           | 0.955 | 134.05               | 0.953 | 170.78          | 0.960 |
|                           |                   |                  |       |                      |       |                 |       |
| Bio-tar                   | 20                | 113.67           | 0.995 | 130.61               | 0.992 | 116.10          | 0.995 |
|                           | 30                | 113.94           | 0.999 | 134.04               | 1.000 | 116.66          | 0.999 |
|                           | 40                | 123.44           | 0.997 | 135.43               | 0.995 | 126.03          | 0.997 |
|                           | 50                | 138.20           | 0.991 | 136.04               | 0.996 | 210.05          | 0.992 |
|                           | 60                | 159.45           | 0.989 | 147.63               | 0.997 | 161.04          | 0.991 |
|                           | 70                | 228.51           | 0.981 | 190.24               | 1.000 | 227.20          | 0.983 |
|                           | 80                | 368.79           | 0.910 | 333.55               | 0.906 | 361.28          | 0.915 |
|                           |                   |                  |       |                      |       |                 |       |
| Bio-tar: Corn stalk (1:3) | 20                | 111.08           | 0.940 | 108.31               | 0.986 | 113.58          | 0.948 |
|                           | 30                | 150.68           | 0.961 | 167.93               | 0.920 | 151.72          | 0.965 |
|                           | 40                | 170.00           | 0.984 | 163.43               | 0.959 | 170.48          | 0.986 |
|                           | 50                | 197.89           | 0.990 | 195.55               | 0.939 | 197.36          | 0.991 |
|                           | 60                | 192.15           | 1.000 | 229.28               | 0.982 | 192.18          | 1.000 |
|                           | 70                | 156.48           | 0.954 | 192.44               | 0.997 | 158.52          | 0.959 |
|                           | 80                | 212.56           | 0.905 | 208.96               | 0.818 | 212.35          | 0.913 |
|                           |                   |                  |       |                      |       |                 |       |

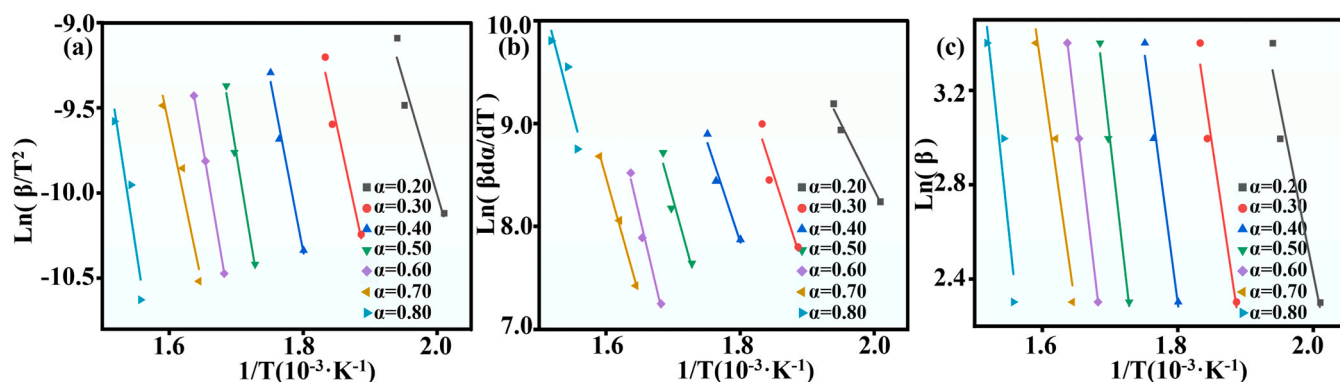


Fig. 5. Comparison of 1:3 mixed samples of bio-tar corn stalk under different models (a. DAEM, b. Friedman, c. FWO).

conversion rate increased. The fitting performance of these models was satisfactory, with  $R^2$  values exceeding 0.950.

Fig. 6 illustrates the activation energy differences for the 1:3 mixture of bio-tar and corn stalk under the influence of different models. From the data presented in the graph, it was evident that the experimental values, under the action of all three models, were significantly lower than the theoretical values. Particularly, under the DAEM model, the activation energy difference reached 72.87 kJ/mol at a conversion rate of 70%. Similarly, under the Friedman and FWO models, the maximum activation energy differences occurred at conversion rates of 40% and 30%, reaching 49.93 kJ/mol and 28.48 kJ/mol, respectively. This further indicates that under the interaction effect, lower activation energy was required to initiate the pyrolysis reaction of the mixed sample. Therefore, the DAEM, Friedman, and FWO models effectively demonstrate the facilitating relationship between the small amount of bio-tar blending and the pyrolytic behavior of corn stalk, highlighting the reasons behind the interaction effect from the perspective of activation energy and revealing the underlying mechanisms governing the occurrence of the reaction in terms of energy input. These findings provide a theoretical basis for understanding the subsequent impacts on the distribution of pyrolysis products.

### 3.4. Product distributions analysis

The pyrolysis experiments were conducted on five different samples under nitrogen atmosphere at 600 °C to determine the distribution of

decomposition products. The results revealed that individual thermal decomposition of corn stalk achieved the highest liquid yield at 48.00%, while bio-tar exhibited the highest gas yield at 54.44%. This indicates a higher content of light volatile components, primarily aromatic compounds with 1–5 benzene rings, in bio-tar, which readily transformed into small, non-condensable gaseous forms under the thermal conditions. In contrast, corn stalk, with its higher content of hemicellulose, cellulose, and lignin, exhibited a greater propensity for conversion into condensable decomposition products such as levoglucosan, ketones, and phenols. Thus, individual corn stalk pyrolysis resulted in the highest oil yield. Under mixed proportion conditions, the 1:1 ratio had the highest liquid yield of 33.83%, while the 1:3 ratio had the highest gas yield of 50.00%. This suggests that the interaction between feedstocks led to increasing gas yield compared to the individual thermal decomposition of bio-tar, accompanied by a decrease in liquid yield. Co-pyrolysis favored the generation of small, non-condensable fractions from volatile components and potentially promoted the secondary cracking of light oil components, resulting in reduced oil yield and increasing gas yield. In conclusion, the distribution of pyrolysis products was significantly influenced by the mixed proportion of different samples. Table 4.

The distribution of liquid products was compared and analyzed in Fig. 7, categorizing them into phenols, hydrocarbons, ketones, furans, and other compound types. The phenol content remained similar under separate thermal decomposition conditions for bio-tar and corn stalk. However, a significant increase to 71.97% in phenol content was observed under the bio-tar:corn stalk ratio of 1:3, highlighting the influence of a small amount of bio-tar addition in promoting phenolic compound enrichment. It should be noted that the effect was not directly proportional to the bio-tar content, as the phenol content decreased to 60.27% and 58.03% under the bio-tar : corn stalk ratios of 1:1 and 3:1, respectively. Conversely, the highest hydrocarbon content of 21.05% was achieved under the bio-tar to corn stalks ratio of 3:1, indicating favorable conditions for hydrocarbon accumulation, potentially involving methoxy group removal and cleavage of other oxygen-containing functional groups. Ketones and furans, the primary products of cellulose and hemicellulose pyrolysis, exhibited higher production levels at the bio-tar to corn stalks ratio of 1:1, with furans reaching 8.44% and ketones reaching 14.26%. The distribution selectivity of others of biooils in the three mixing ratios was similar. Thus, a lower proportion of bio-tar addition promoted phenolic compound generation,

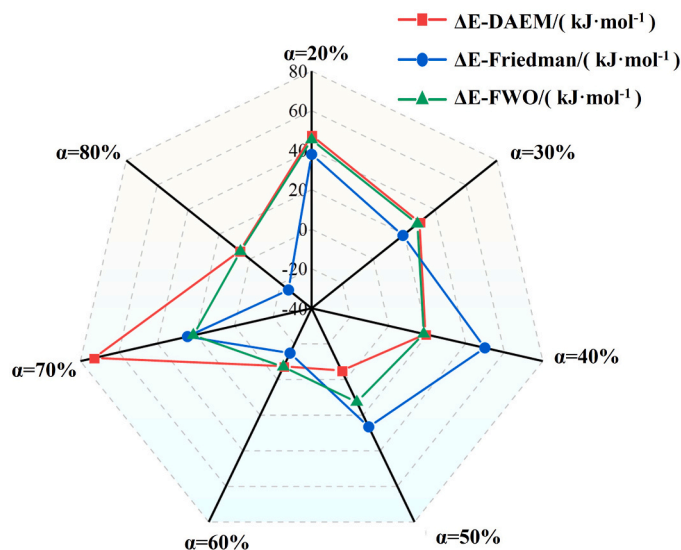


Fig. 6. The difference of activation energy of 1:3 of bio-tar:corn stalk under different models (calculated value-experimental value).

Table 4

Comparison of pyrolysis three-state product yields of different samples.

| Sample                    | Biochar/% | Biooil/% | Gases/% |
|---------------------------|-----------|----------|---------|
| Bio-tar: Corn stalk = 1:0 | 25.56     | 20.00    | 54.44   |
| Bio-tar: Corn stalk = 3:1 | 34.67     | 25.63    | 39.70   |
| Bio-tar: Corn stalk = 1:1 | 27.86     | 33.83    | 38.31   |
| Bio-tar: Corn stalk = 1:3 | 26.49     | 37.36    | 36.15   |
| Bio-tar: Corn stalk = 0:1 | 25.00     | 48.00    | 27.00   |

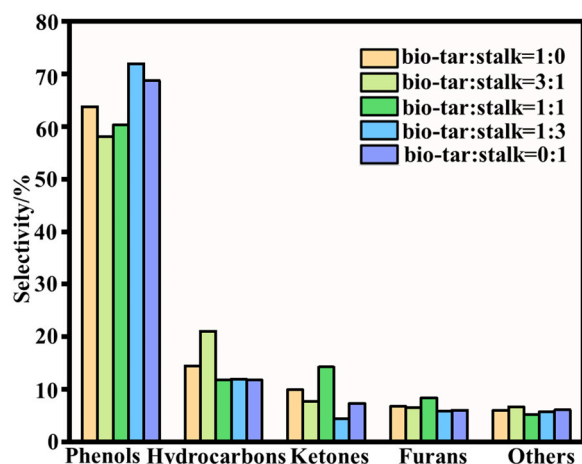


Fig. 7. Distribution of liquid products in different samples (others contains acids, esters, and nitrogen compounds).

a moderate bio-tar addition favored hydrocarbon formation, and a higher bio-tar content enhanced the synthesis of ketones and furans.

#### 4. Conclusions

The present study investigates the co-pyrolysis of corn stalk and bio-tar to explore their interaction and provides insights for the utilization of these two important biomass resources. The research primarily focuses on the thermal behavior analysis through thermogravimetric analysis, kinetic modeling using Distributed Activation Energy (DAEM), Friedman, Kissenger-Akahira-Sunose (KAS) and Flynn-Wall-Ozawa (FWO) models, and analysis of product distribution during co-pyrolysis. The experimental results reveal that there exists a significant interaction between bio-tar and corn stalk at different mixing ratios. Specifically, the co-pyrolysis of bio-tar and corn stalk with a ratio of 1:3 exhibits the most pronounced synergistic effect, promoting the thermal decomposition of corn stalk. The kinetic modeling calculation demonstrates a reduction in activation energy for all mixing ratios, with a significant decrease of 72.87 kJ/mol observed for the bio-tar:stalk ratio of 1:3. Furthermore, the distribution of pyrolysis products varies with different mixing ratios, where in a lower bio-tar ratio favors the production of phenolic compounds, an intermediate ratio enhances the formation of hydrocarbons, and a higher bio-tar ratio promotes the synthesis of ketones and furans. Overall, this research advances the understanding of the interaction between bio-tar and corn stalk during co-pyrolysis and provides valuable insights for the efficient utilization of biomass resources.

#### CRediT authorship contribution statement

**Chen Yuhuan:** Software, Investigation, Formal analysis. **Xie Geliang:** Methodology, Investigation, Conceptualization. **Hu Ming:** Writing – review & editing, Writing – original draft, Data curation. **Zhu Guo-qiang:** Writing – original draft, Methodology, Investigation, Data curation. **Xu Lujiang:** Writing – review & editing, Writing – original draft, Supervision, Funding acquisition. **Zhu Mingxu:** Visualization, Data curation, Conceptualization. **Lv Tong:** Validation, Software.

#### Declaration of Competing Interest

The authors declared that there is no conflict of interest.

#### Data availability

Data will be made available on request.

#### Acknowledgements

The authors are grateful to the financial supports from, the National Natural Science Foundation of China (52376196), Natural Science Foundation of Jiangsu Province (BK20231477), “Innovation & Entrepreneurship Talents” Introduction Plan of Jiangsu Province.

#### References

- [1] H. Zhang, K. Yang, Y. Tao, Q. Yang, L. Xu, C. Liu, L. Ma, R. Xiao, Biomass directional pyrolysis based on element economy to produce high-quality fuels, chemicals, carbon materials – A review, *Biotechnol. Adv.* 69 (2023) 108262, <https://doi.org/10.1016/j.biotechadv.2023.108262>.
- [2] A. Bohre, A. Modak, V. Chourasia, P. Ram Jadhao, K. Sharma, K. Kishore Pant, Recent advances in supported ionic liquid catalysts for sustainable biomass valorisation to high-value chemicals and fuels, *Chem. Eng. J.* 450 (2022) 138032, <https://doi.org/10.1016/j.cej.2022.138032>.
- [3] S. Wang, A. Cheng, F. Liu, J. Zhang, T. Xia, X. Zeng, W. Fan, Y. Zhang, Catalytic conversion network for lignocellulosic biomass valorization: a panoramic view, *Ind. Chem. Mater.* 1 (2) (2023) 188–206, <https://doi.org/10.1039/D2IM00054G>.
- [4] J. Lee, K.-H. Kim, E.E. Kwon, Biochar as a Catalyst, *Renew. Sust. Energ. Rev.* 77 (2017) 70–79, <https://doi.org/10.1016/j.rser.2017.04.002>.
- [5] L. Leng, X. Xu, L. Wei, L. Fan, H. Huang, J. Li, Q. Lu, J. Li, W. Zhou, Biochar stability assessment by incubation and modelling: Methods, drawbacks and recommendations, *Sci. Total Environ.* 664 (2019) 11–23, <https://doi.org/10.1016/j.scitotenv.2019.01.298>.
- [6] Y. Xie, L. Wang, H. Li, L.J. Westholm, L. Carvalho, E. Thorin, Z. Yu, X. Yu, O. Skreiberg, A critical review on production, modification and utilization of biochar, *J. Anal. Appl. Pyrol.* 161 (2022), <https://doi.org/10.1016/j.jaap.2021.105405>.
- [7] K. Luo, C. Zhang, S. Zhu, Y. Bai, F. Li, Tar formation during coal pyrolysis under N<sub>2</sub> and CO<sub>2</sub> atmospheres at elevated pressures, *J. Anal. Appl. Pyrol.* 118 (2016) 130–135, <https://doi.org/10.1016/j.jaap.2016.01.009>.
- [8] S. Zhang, Q. Yao, H. Pan, Y. Liu, Z. Shen, Z. Ge, L. Li, Y. Zhang, X. Li, Y. Wang, Highly efficient steam reforming of biomass tar model compound using a high-entropy alloy-based structured catalyst, *Int. J. Hydrog. Energy* (2023), <https://doi.org/10.1016/j.ijhydene.2023.10.315>.
- [9] N. Gao, J. Salisu, C. Quan, P. Williams, Modified nickel-based catalysts for improved steam reforming of biomass tar: A critical review, *Renew. Sustain. Energy Rev.* 145 (2021) 111023, <https://doi.org/10.1016/j.rser.2021.111023>.
- [10] J. Li, J. Tao, B. Yan, L. Jiao, G. Chen, J. Hu, Review of microwave-based treatments of biomass gasification tar, *Renew. Sustain. Energy Rev.* 150 (2021) 111510, <https://doi.org/10.1016/j.rser.2021.111510>.
- [11] Y. Shen, Y. Fu, Advances in in situ and ex situ tar reforming with biochar catalysts for clean energy production, *Sustain. Energy. Fuels* 2 (2) (2018) 326–344, <https://doi.org/10.1039/c7se00553a>.
- [12] J. Li, D. Yang, X. Yao, H. Zhou, K. Xu, L. Geng, Slow pyrolysis experimental investigation of biomass tar formation and hydrogen production by tar reforming, *Int. J. Hydrog. Energy* (2023), <https://doi.org/10.1016/j.ijhydene.2023.01.326>.
- [13] M. Zhang, G. Fan, N. Liu, M. Yang, X. Li, Y. Wu, Tar removal in pine pyrolysis catalyzed by bio-char supported nickel catalyst, *J. Anal. Appl. Pyrolysis* 169 (2023) 105843, <https://doi.org/10.1016/j.jaap.2022.105843>.
- [14] A. Fourcault, F. Marias, U. Michon, Modelling of thermal removal of tars in a high temperature stage fed by a plasma torch, *Biomass-- Bioenergy* 34 (9) (2010) 1363–1374, <https://doi.org/10.1016/j.biombioe.2010.04.018>.
- [15] N. Gao, M.H. Milandile, C. Quan, L. Rundong, Critical assessment of plasma tar reforming during biomass gasification: A review on advancement in plasma technology, *J. Hazard. Mater.* 421 (2022) 126764, <https://doi.org/10.1016/j.jhazmat.2021.126764>.
- [16] W. Ma, C. Chu, P. Wang, Z. Guo, B. Liu, G. Chen, Characterization of tar evolution during DC thermal plasma steam gasification from biomass and plastic mixtures: Parametric optimization via response surface methodology, *Energy Convers. Manag.* 225 (2020) 113407, <https://doi.org/10.1016/j.enconman.2020.113407>.
- [17] J.-H. Lee, I.-G. Lee, J.-Y. Park, K.-Y. Lee, In-situ upgrading of bio-tar over Mg-Ni-Mo catalyst supported by KOH treated activated charcoal in supercritical ethanol, *Fuel* 247 (2019) 334–343, <https://doi.org/10.1016/j.fuel.2019.03.035>.
- [18] S. Vivanpatarakij, S. Assabumrungrat, Thermodynamic analysis of combined unit of biomass gasifier and tar steam reformer for hydrogen production and tar removal, *Int. J. Hydrog. Energy* 38 (10) (2013) 3930–3936, <https://doi.org/10.1016/j.ijhydene.2012.12.039>.
- [19] Y. Zhai, T. Yang, Y. Zhang, Y. Zhu, Co-pyrolysis characteristics of raw/torrefied corn stalk and oil shale, *J. Anal. Appl. Pyrolysis* 171 (2023) 105967, <https://doi.org/10.1016/j.jaap.2023.105967>.
- [20] W. Hong, Y. Zhang, H. Jiang, S. Li, J. Chen, L. Zhang, Co-pyrolysis of corn stalk and coal fly ash: A case study on catalytic pyrolysis behavior, bio-oil yield and its characteristics, *Case Stud. Therm. Eng.* 38 (2022) 102346, <https://doi.org/10.1016/j.csite.2022.102346>.
- [21] H. Sun, D. Feng, S. Sun, Y. Zhao, L. Zhang, G. Chang, Q. Guo, J. Wu, Y. Qin, Effect of acid washing and K/Ca loading on corn straw with the characteristics of gas-solid products during its pyrolysis, *Biomass-- Bioenergy* 165 (2022) 106569, <https://doi.org/10.1016/j.biombioe.2022.106569>.
- [22] M. Xiong, J. Huang, X. He, Z. Zhou, X. Qu, S. Faisal, A. Abomohra, Evaluation of bio-oil/biodiesel production from co-pyrolysis of corn straw and natural hair: A

- new insight towards energy recovery and waste biorefinery, *Fuel* 331 (2023) 125710, <https://doi.org/10.1016/j.fuel.2022.125710>.
- [23] G. Guan, M. Kaewpanha, X. Hao, A. Abudula, Catalytic steam reforming of biomass tar: Prospects and challenges, *Renew. Sustain. Energy Rev.* 58 (2016) 450–461, <https://doi.org/10.1016/j.rser.2015.12.316>.
- [24] G. Montero, M.A. Coronado, R. Torres, B.E. Jaramillo, C. García, M. Stoytcheva, A. M. Vázquez, J.A. León, A.A. Lambert, E. Valenzuela, Higher heating value determination of wheat straw from Baja California, Mexico, *Energy* 109 (2016) 612–619, <https://doi.org/10.1016/j.energy.2016.05.011>.
- [25] J. Zou, H. Hu, Y. Li, H. Jahangiri, F. He, X. Zhang, M.M. Rahman, J. Cai, Complementary use of generalized logistic mixture model and distributed activation energy model in exploring kinetic mechanisms of wheat straw and torrefied rice husk pyrolysis, *J. Clean. Prod.* 397 (2023) 136560, <https://doi.org/10.1016/j.jclepro.2023.136560>.
- [26] Y. Lin, Y. Tian, Y. Xia, S. Fang, Y. Liao, Z. Yu, X. Ma, General distributed activation energy model (G-DAEM) on co-pyrolysis kinetics of bagasse and sewage sludge, *Bioresour. Technol.* 273 (2019) 545–555, <https://doi.org/10.1016/j.biortech.2018.11.051>.
- [27] D. Chen, D. Gao, S. Huang, S.C. Capareda, X. Liu, Y. Wang, T. Zhang, Y. Liu, W. Niu, Influence of acid-washed pretreatment on the pyrolysis of corn straw: A study on characteristics, kinetics and bio-oil composition, *J. Anal. Appl. Pyrolysis* 155 (2021) 105027, <https://doi.org/10.1016/j.jaap.2021.105027>.
- [28] Y. Wang, S. Yang, G. Bao, H. Wang, Investigation of tobacco straw pyrolysis: Three-parallel Gaussian reaction modeling, products analysis and ANN application, *Ind. Crops Prod.* 200 (2023) 116864, <https://doi.org/10.1016/j.indcrop.2023.116864>.
- [29] N. Gautam, A. Chaurasia, Study on kinetics and bio-oil production from rice husk, rice straw, bamboo, sugarcane bagasse and neem bark in a fixed-bed pyrolysis process, *Energy* 190 (2020) 116434, <https://doi.org/10.1016/j.energy.2019.116434>.
- [30] R. Xiao, W. Yang, X. Cong, K. Dong, J. Xu, D. Wang, X. Yang, Thermogravimetric analysis and reaction kinetics of lignocellulosic biomass pyrolysis, *Energy* 201 (2020) 117537, <https://doi.org/10.1016/j.energy.2020.117537>.
- [31] Y. Jiang, C. Li, L. Zhang, S. Zhang, D. Wang, C. Leng, Y. Tang, Z. Cui, X. Hu, Impacts of acidic or basic sites on poplar sawdust on pyrolytic products and pyrolysis kinetics, *J. Anal. Appl. Pyrolysis* 174 (2023) 106151, <https://doi.org/10.1016/j.jaap.2023.106151>.
- [32] Z.B. Laouge, H. Merdun, Investigation of thermal behavior of pine sawdust and coal during co-pyrolysis and co-combustion, *Energy* 231 (2021) 120895, <https://doi.org/10.1016/j.energy.2021.120895>.
- [33] S. Huang, Y. Su, W. Luo, Q. He, S. Huang, N. Zhou, Z. Zhou, Kinetic analysis and in-situ no support catalytic pyrolysis product distribution of Chinese herb residue, *J. Anal. Appl. Pyrolysis* 156 (2021) 105114, <https://doi.org/10.1016/j.jaap.2021.105114>.
- [34] S.B. Prajapati, A. Gautam, S. Gautam, Non-isothermal kinetic study by TGA analysis of printed circuit boards and cotton stalk, *Biomass-Bioenergy* 172 (2023) 106746, <https://doi.org/10.1016/j.biombioe.2023.106746>.
- [35] P. Ling, J. Xu, T. Liu, X. An, X. Wang, M.E. Mostafa, H. Han, K. Xu, L. Jiang, Y. Wang, S. Su, S. Hu, J. Xiang, Pyrolysis kinetics and reaction mechanisms of coal slime for cleaner energy, *J. Anal. Appl. Pyrolysis* 168 (2022) 105718, <https://doi.org/10.1016/j.jaap.2022.105718>.
- [36] Y. Li, H. Zhao, X. Sui, X. Wang, H. Ji, Studies on individual pyrolysis and co-pyrolysis of peat-biomass blends: Thermal decomposition behavior, possible synergism, product characteristic evaluations and kinetics, *Fuel* 310 (2022) 122280, <https://doi.org/10.1016/j.fuel.2021.122280>.
- [37] S. Fang, Z. Yu, Y. Lin, S. Hu, Y. Liao, X. Ma, Thermogravimetric analysis of the co-pyrolysis of paper sludge and municipal solid waste, *Energy Convers. Manag.* 101 (2015) 626–631, <https://doi.org/10.1016/j.enconman.2015.06.026>.
- [38] Y. Guan, C. Liu, Q. Peng, F. Zaman, H. Zhang, Z. Jin, A. Wang, W. Wang, Y. Huang, Pyrolysis kinetics behavior of solid leather wastes, *Waste Manag.* 100 (2019) 122–127, <https://doi.org/10.1016/j.wasman.2019.09.005>.
- [39] J. Kristanto, A.F. Daniyal, D.Y. Pratama, I.N.M. Bening, L. Setiawan, M.M. Azis, S. Purwono, Kinetic Study on the slow pyrolysis of isolated cellulose and lignin from teak sawdust, *Thermochim. Acta* 711 (2022) 179202, <https://doi.org/10.1016/j.tca.2022.179202>.
- [40] R. Kumar Mishra, K. Mohanty, Kinetic analysis and pyrolysis behavior of low-value waste lignocellulosic biomass for its bioenergy potential using thermogravimetric analyzer, *Mater. Sci. Energy Technol.* 4 (2021) 136–147, <https://doi.org/10.1016/j.mset.2021.03.003>.
- [41] R.N. Mandapati, P.K. Ghodke, Kinetics of pyrolysis of cotton stalk using model-fitting and model-free methods, *Fuel* 303 (2021) 121285, <https://doi.org/10.1016/j.fuel.2021.121285>.
- [42] S. Pattanayak, L. Hauchhum, C. Loha, L. Sailo, L. Mishra, Experimental investigation on pyrolysis kinetics, reaction mechanisms and thermodynamic parameters of biomass and tar in N<sub>2</sub> atmosphere, *Sustain. Energy Technol. Assess.* 48 (2021) 101632, <https://doi.org/10.1016/j.seta.2021.101632>.
- [43] T. Len, V. Bressi, A.M. Balu, T. Kulik, O. Korchuganova, B. Palianytsia, C. Espro, R. Luque, Thermokinetics of production of biochar from crop residues: an overview, *Green. Chem.* 24 (20) (2022) 7801–7817, <https://doi.org/10.1039/d2gc02631g>.
- [44] A.N. Amenaghawon, C.L. Anyalewechi, C.O. Okieimen, H.S. Kusuma, Biomass pyrolysis technologies for value-added products: a state-of-the-art review, *Environ., Dev. Sustain.* 23 (10) (2021) 14324–14378, <https://doi.org/10.1007/s10668-021-01276-5>.